AUTOCAD

MINOR PROJECT-AUTOCAD TRAINEESHIP PROGRAM

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INTRODUCTION

- AutoCAD is a **Computer-Aided Design (CAD)** software developed by *Autodesk Inc.*

- It allows engineers, architects, and designers to **create precise 2D and 3D drawings**.

- Used across multiple disciplines: Civil, Mechanical, Electrical, and Architecture.

- Despite the rise of **Building Information Modeling (BIM)** software like Revit or **Parametric Modeling** like SolidWorks, AutoCAD remains the "universal language" of the design world. Most project handovers and site plans are still exchanged as .dwg files because of their unmatched precision in 2D documentation.

WHAT IS AUTOCAD

- Launched in **1982**.
- Before 1982, engineers used "drafting tables" and pencils. AutoCAD revolutionized the industry by allowing **reusability** (copy/paste).
- “AutoCAD” stands for **Automatic Computer-Aided Design**.
- Supports **drafting, modeling, rendering, and documentation**.
- Integrates with **AI and cloud tools** for improved design workflow.
- Unlike a photo (pixels), AutoCAD uses mathematical vectors. This means you can zoom into a bolt on a spaceship or zoom out to see the whole station without losing any clarity.

USES OF AUTOCAD

- Floor plans, electrical layouts, and plumbing diagrams are examples of 2D drafting.
- 3D modeling includes industrial tools, architectural models, and mechanical components.Measurement, annotation, and standardized reporting are examples of documentation.
- Visualization comprises isometric drawings, sectional views, and realistic representation.

APPLICATION AREAS

Field	Usage Example
Civil	Structural plans, roads, bridges
Mechanical	Gears, machine parts
Electrical	Wiring layouts
Architecture	Building layouts, interiors
Aerospace	Components, assemblies

Mention that AutoCAD is used for schematics (wiring/piping) and first layouts in the Mechanical/Aerospace area before switching to more sophisticated 3D applications like Inventor or SolidWorks.

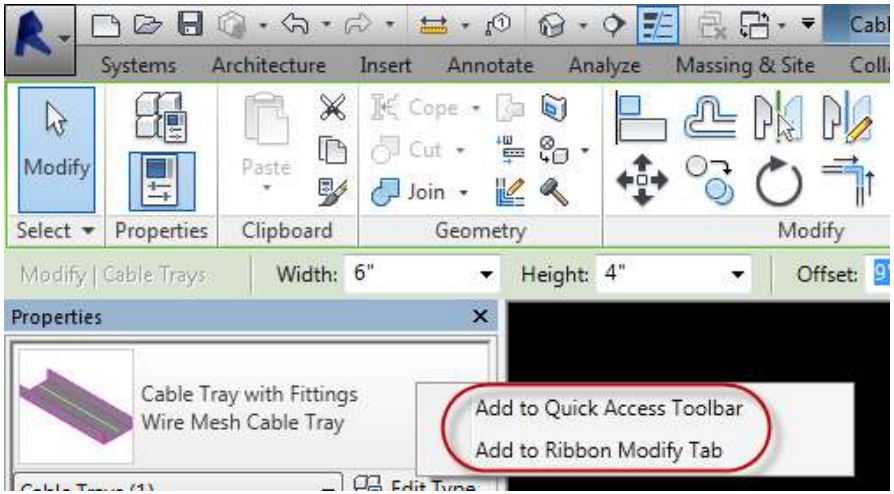
AUTOCAD INTERFACE OVERVIEW

INTERFACE INTRODUCTION

- Key interface elements:
  - Ribbon
  - Toolbars
  - Command Line
  - Drawing Area
  - Status Bar
  - Navigation Cube

RIBBON

- located at the very top.
- includes panels and tabs such as Home, Insert, Annotate, and View.
- presents a collection of drawing and editing tools.
- It's contextual. The Ribbon immediately enters "Text Editor" mode when you click on a text passage.



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COMMAND LINE

- positioned at the base.
- enables access of text-based commands.
- Examples are MOVE, CIRCLE, and LINE.
- You write L for line or C for circle instead of looking for icons. Additionally, it offers suggestions; for example, the sentence "Specify center point" demonstrates you exactly what to do once you start a command.

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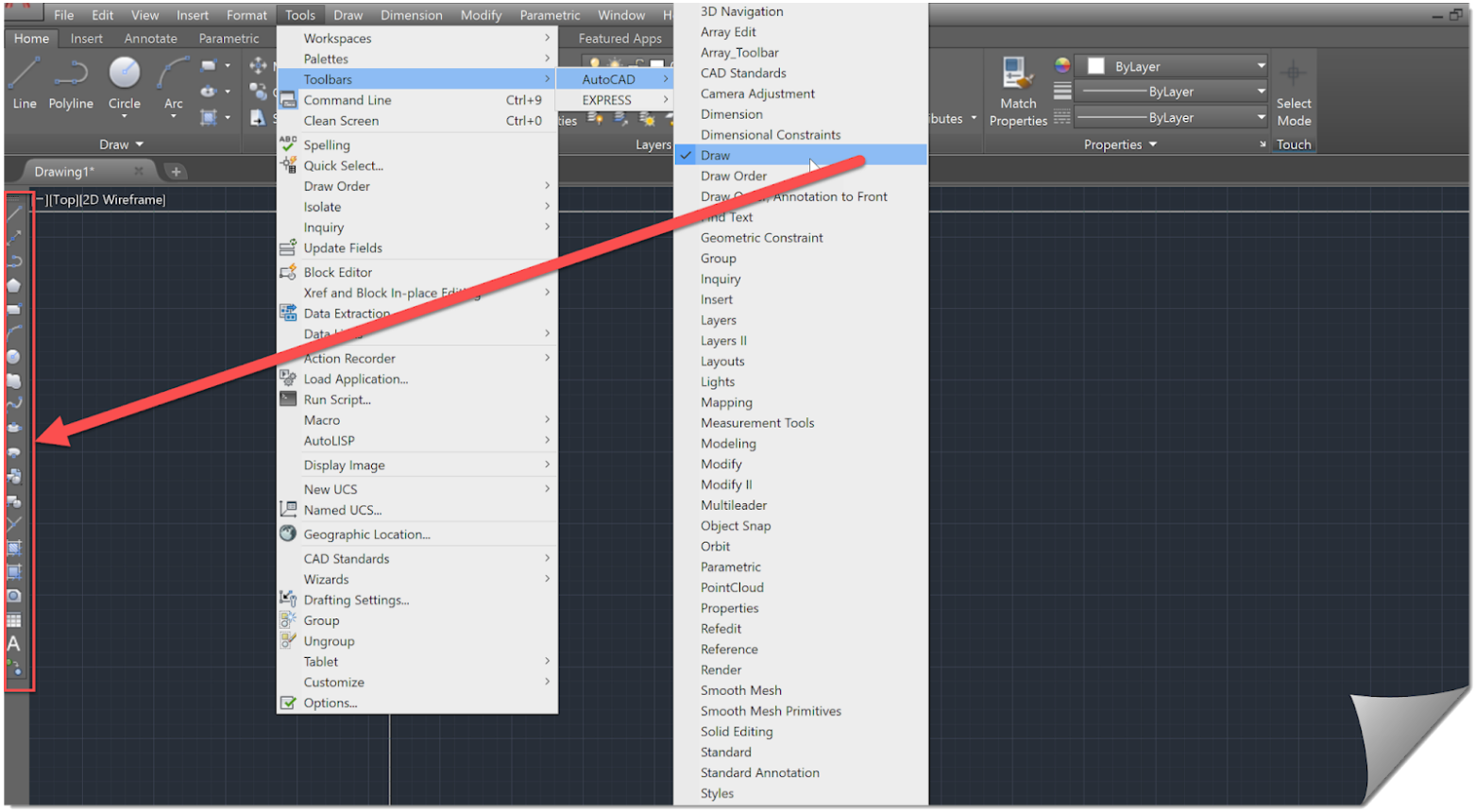
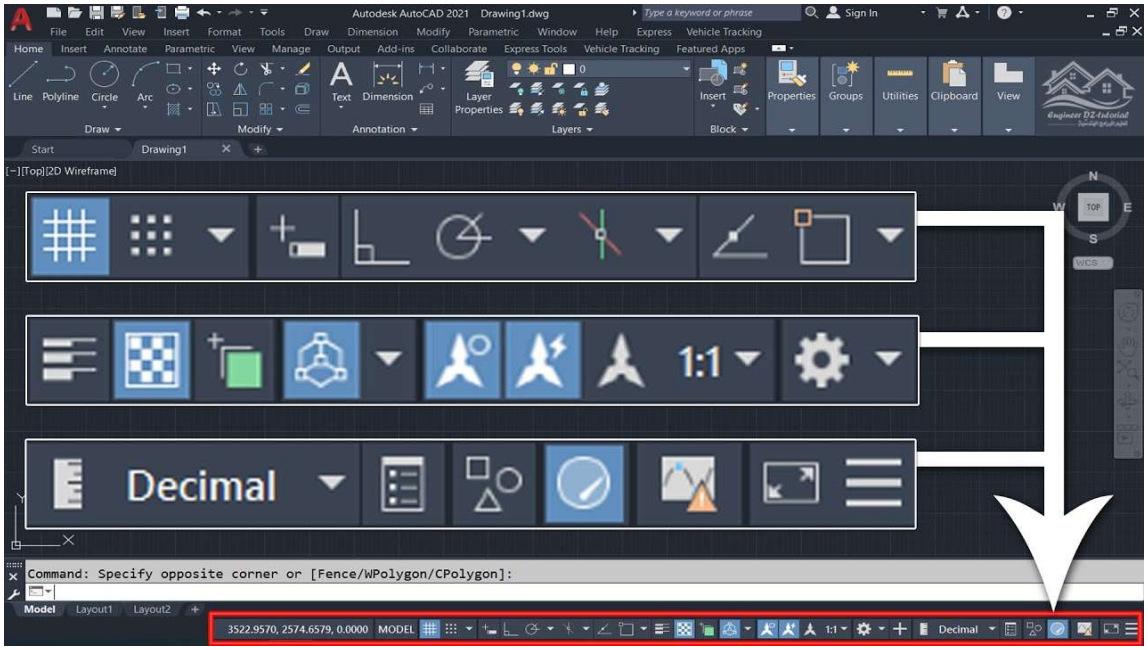
WORKSPACE AND TOOLBAR

- Workspaces: 3D Modeling, 3D Basics, and Drafting & Annotation.
- For frequently used commands, toolbars can be changed.



STATUS BAR

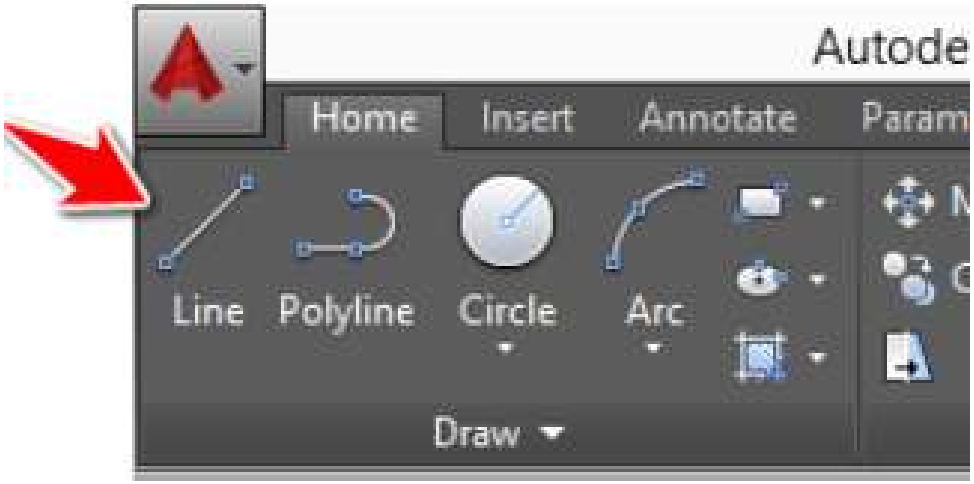
- displays graphics tools like:
- Grid, Snap, Ortho, Object Snap, Polar Tracking, and Dynamic Input.
- aids in keeping drawings accurate.



BASIC DRAWING COMMANDS

LINE COMMAND

- draws lines which are straight between two points.
- Command: LINE or the shortcut L + Enter.
- Use: Technical outlines, walls, and edges.s



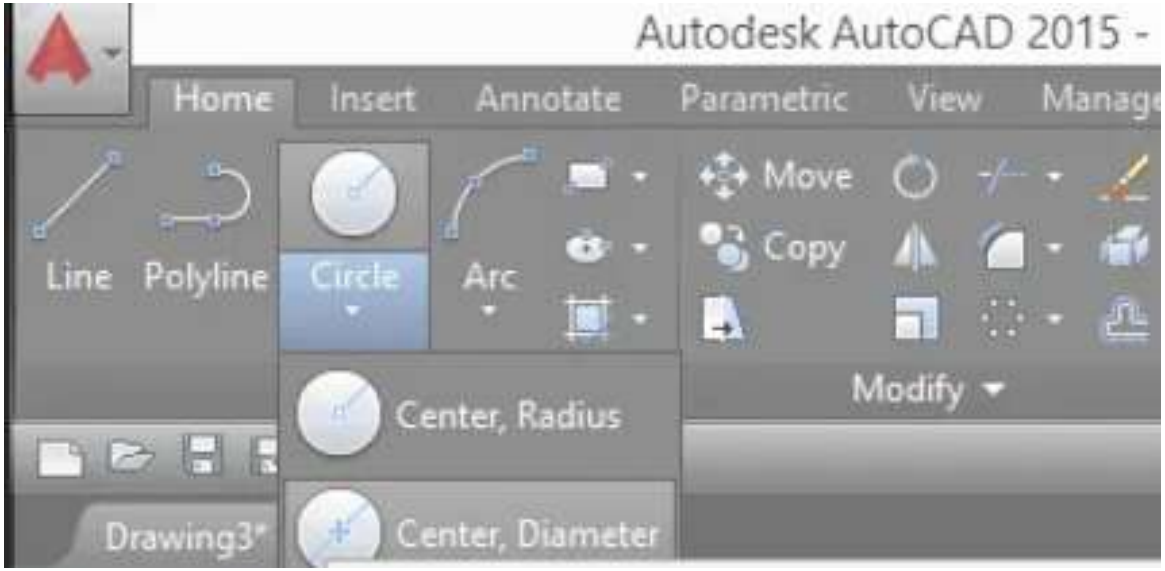
CIRCLE COMMAND

renders a circle with:

1)Center and Diameter

2)Center and Radius

Command: C + Enter or CIRCLE.



ARC AND RECTANGLE

- ARC: forms a segment that is curved. (ARC)
- RECTANGLE: Draws shapes with four sides. (RECTANG)
- is used in structural drawings or mechanical corners.

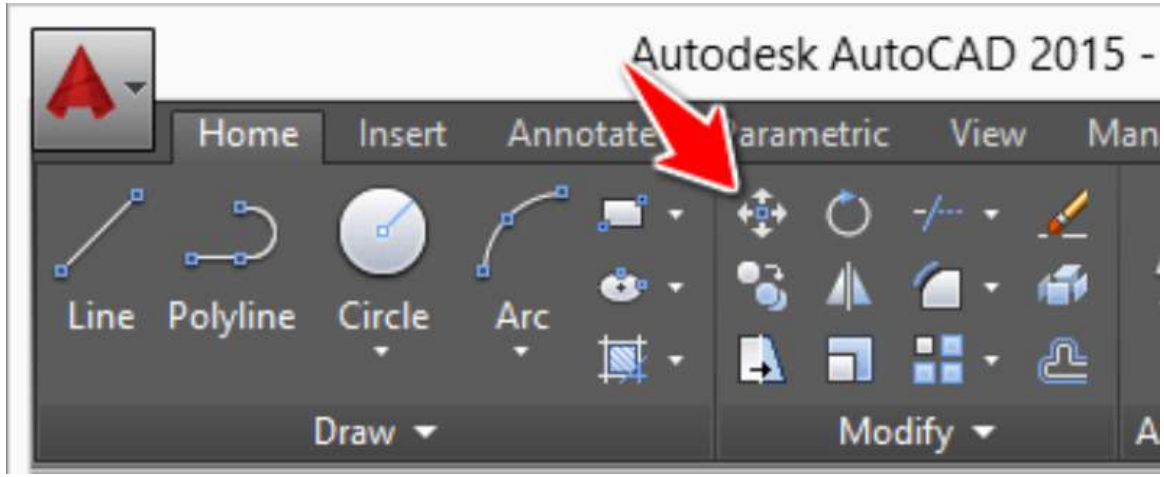
POLYGON & ELLIPSE

- Polygons are regular, closed shapes with equal sides. (Polygon)
- ELLIPSE: Forms oval forms. (ELLIPSE)
- AutoCAD asks you if you want the shape to be circumscribed (outside a circle) or inscribed (within a circle). For mechanical components like hexagonal bolt heads, this is essential.

MODIFY COMMANDS

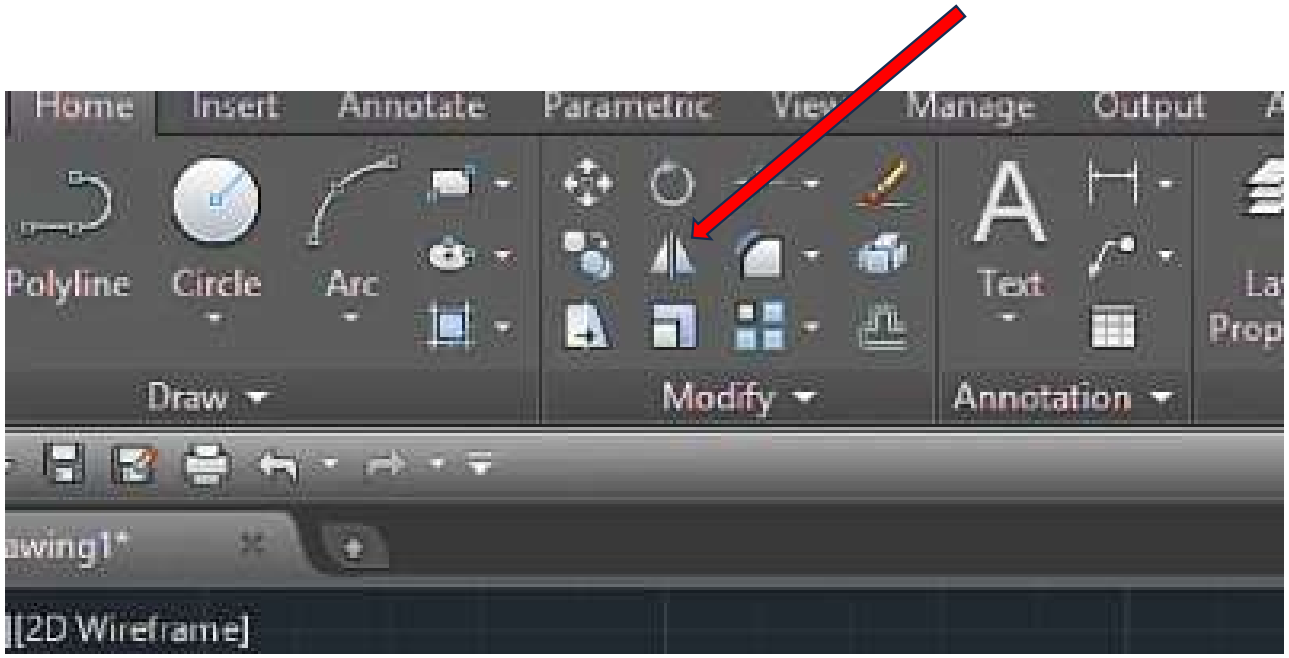
MOVE AND COPY

- MOVE: Moves objects from one spot to another. (MOVE)
- COPY: Duplicates specific items. (COPY)



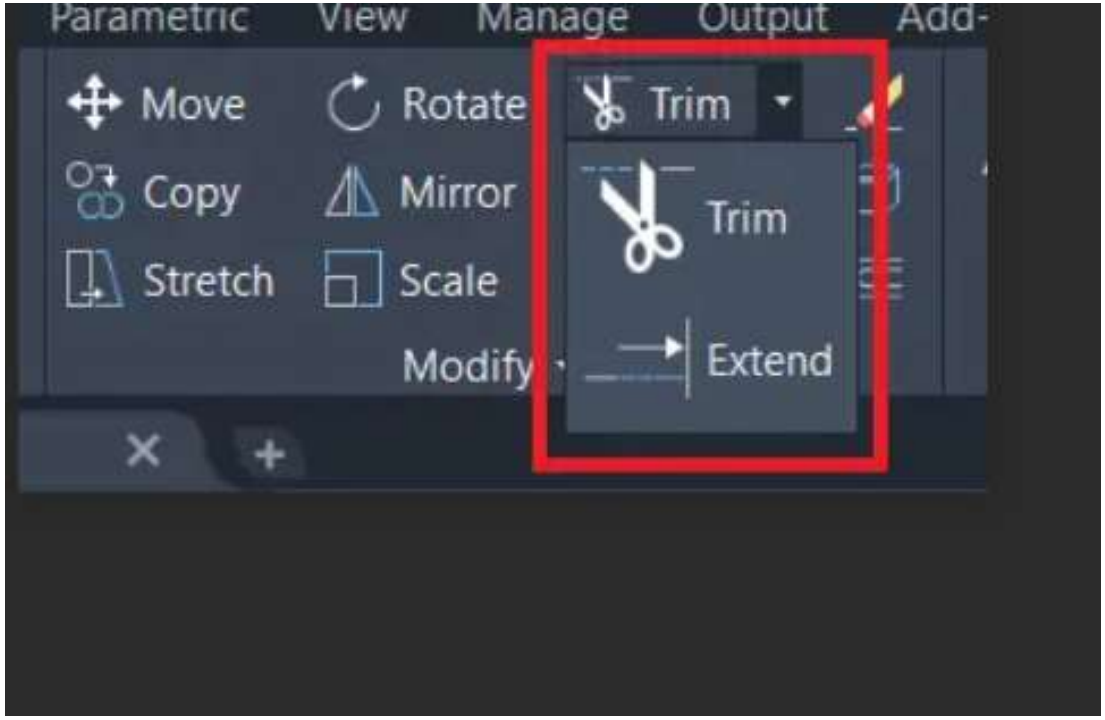
ROTATE & MIRROR

- *ROTATE: Revolves items around a starting position. (ROTATE)*
- *MIRROR: Produces a copy that is reflected across an axis. (MIRROR)*



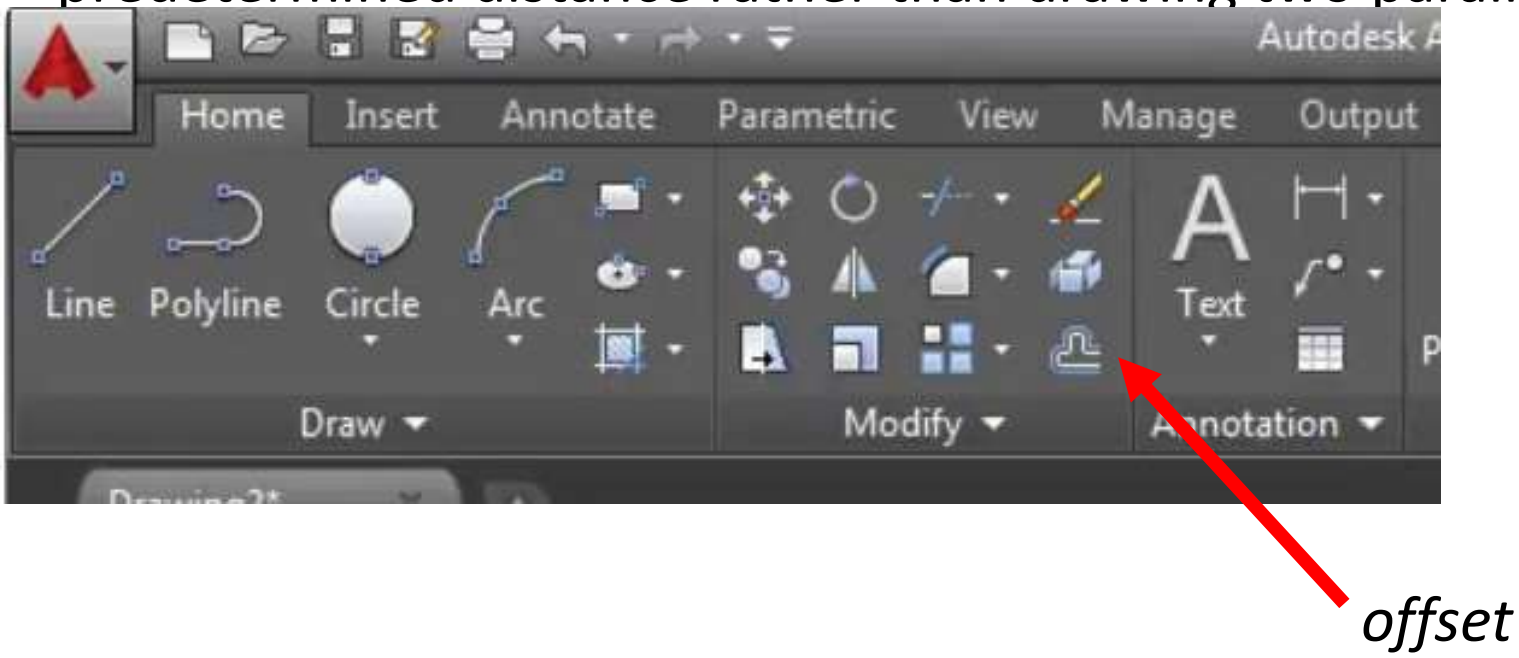
TRIM & EXTEND

- TRIM: Cuts items to align with other objects' edges.
- EXTEND: Extends the lines of an object to their edges.
- Commands: TRIM/EXTEND.



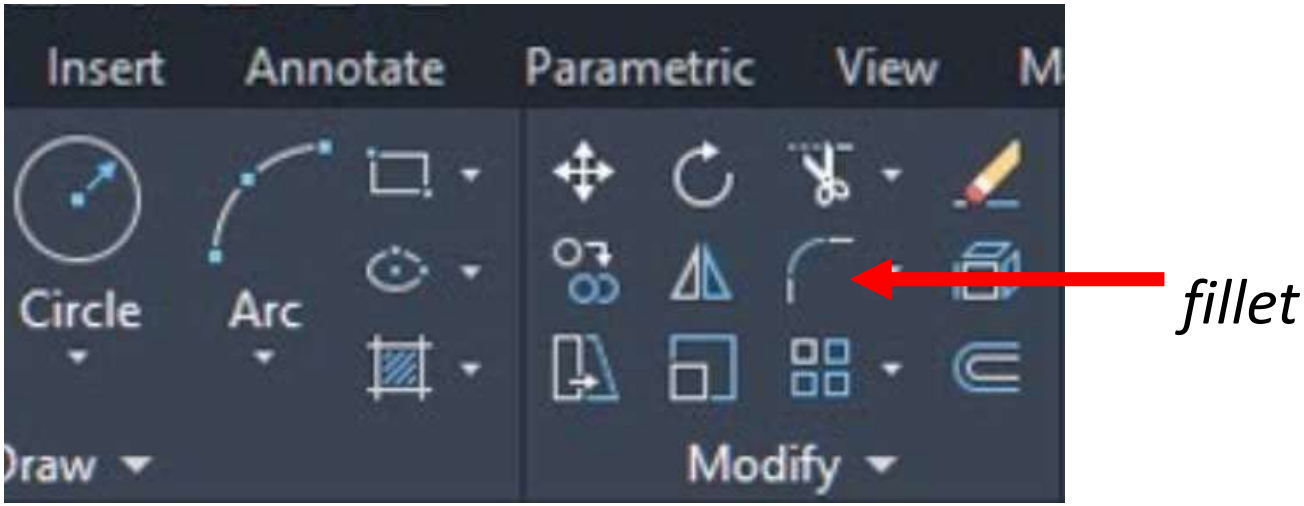
OFFSET, SCALE

- OFFSET: Produces copies in parallel. (OFFSET)
- SCALE: Modifies an object's size without changing its proportions. (Scale)
- For a wall or pipe, you draw one line and "Offset" it by a predetermined distance rather than drawing two parallel lines.



FILLET & CHAMFER

- FILLET: Rounds corners. (Fillet)
- CHAMFER: Angle-bevels corners. CHAMFERFillet aids in lowering the percentages of stress in mechanical components.
- In 2D and 3D modeling, chamfers—sloped or angled transitions between two edges or surfaces—are frequently utilized to enhance assembly fit, safety, and aesthetics.



KEYBOARD SHORTCUTS

Common Shortcut Keys

Command	Shortcut
Line	L
Circle	C
Move	M
Copy	CO
Trim	TR
Offset	O
Zoom	Z
Erase	E

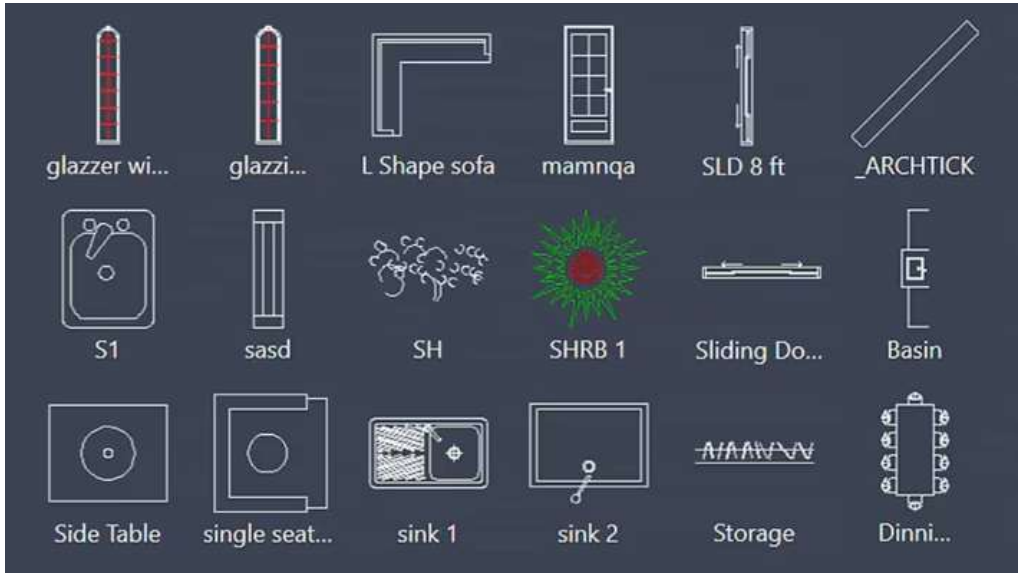
Productivity Shortcuts

Action	Shortcut
Save Drawing	Ctrl + S
Open File	Ctrl + O
Undo	Ctrl + Z
Redo	Ctrl + Y
Zoom Extents	ZE
Regen All	REA

BLOCKS AND ANNOTATIONS

What Are Blocks?

- A collection of items combined into a single symbol is called a block.
- provides consistency in recurring design components and saves time.(e.g., doors, bolts).



Creating and Inserting a Block

Creating a block:

- Launch AutoCAD and open your drawing.
- Locate the Block panel by selecting the Home tab.
- Press Enter after typing BLOCK into the command line or clicking the Create icon.
- Give the block a name in the Name area of the Block Definition dialog box. Click Pick Point and choose a point in the drawing to specify a base point. Click Select Objects and select the objects you want to include in the block in the drawing area. If necessary, select more options like Convert to Block.
- To construct the block, click OK.

Inserting a Block:

- Open the **Blocks Palette** by typing BLOCKSPALETTE in the command line or selecting it from the **Insert** tab.
- Choose a block from the **Current Drawing**, **Recent**, or **Libraries** tab.
- Move your pointer to the sketching area by clicking on the block.
- As necessary, indicate the Rotation, Scale, and Insertion Point.
- To position the block, click.

Renaming a Block:

- Type RENAME in the command line and press Enter.
- Choose Blocks from the list in the Rename dialog box.
- Select the block you wish to rename, then fill in the Rename To field with the new name.
- To implement the changes, click OK.

Deleting a Block

Use Select and Delete to remove every instance of the block in the drawing.
In the command line, type PURGE and hit Enter.
Choose the block you wish to delete from the Block Table in the Purge dialog box.
To remove the block definition, click Purge.

Accessing the Pre-Built Blocks

Go to the Libraries tab after opening the Blocks Palette.
Use Autodesk's Online Block Library by choosing Get or connect to a folder.
Additional Blocks from the Insert tab.
You can browse and add blocks to your drawing directly.

Text and Multiline Text

TEXT: For notes with just one line.
MTEXT: For paragraphs with more than one line.
Tools can be found under the Annotate tab.
To produce text:
Either type TEXT into the command line or choose Single Line Text from the Ribbon (Annotate > Text > Single Line Text) to activate the TEXT command.
1 Indicate where the text should begin.
Using your drawing scale as a guide, enter the text height.
For horizontal text, set the rotation angle to 0; for angled text, use any value.

Dimensions and Styles

- Types: Leader, Diameter, Angular, Radial, Aligned, and Linear.
- To manage dimension styles, use the command DIMSTYLE.
- Make sure the measurements are clear.
- It's about communication, not simply lines. Whether printing on an A4 sheet or a large blueprint, DIMSTYLE lets you adjust the scale so the numbers are readable.

3D DRAWING SETUP

Switching to 3D Workspace

- Choose “3d Basics” or “3d modelling” as your workspace
- tools: Navigation bar,viewcube and visual styles

Basic 3D Commands:

Command	Function
EXTRUDE	Converts 2D profiles into 3D objects
REVOLVE	Rotates 2D shape around axis
UNION	Joins solids
SUBTRACT	Cuts one solid from another

View Controls

Mastering 3D view controls in AutoCAD enhances model inspectio
n,
design accuracy, and workflow efficiency. Key tools include **ViewCube, Orbit, SteeringWheels, Pan/Zoom**, and **Visual Styles**. Combining these controls with mouse and keyboard shortcuts creates smooth navigation and better understanding of the 3D workspace.

- **Orbit:** Rotate view in 3D. (3DORBIT)
- **Viewcube:** Orient models quickly.
- Display modes: Wireframe, Realistic, Shaded.
- **Extrude:** Taking a 2D shape (like a circle) and giving it "thickness" to make a cylinder.
- **Boolean Operations (Union/Subtract):** This is how you "drill" holes in 3D. You create a solid cylinder, place it inside a block, and use the **SUBTRACT** command to remove that volume.

CONCLUSION

pointing out that although software evolves, geometry's logic doesn’t.
It is considerably simpler to learn any other engineering program after you have mastered basic AutoCAD
commands.
For engineers and designers, AutoCAD is a strong and adaptable tool.
A solid drafting foundation is created by comprehending the interface, commands, and 3D tools.
The secret to mastering AutoCAD is "precision and practice."